

Principles of Economics

Problem Set 1

Universidad Carlos III de Madrid | Department of Economics

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1. WARMING-UP QUESTIONS

1. Use one of the following figures to answer the questions paying attention to the difference between the conventional and the ratio scale or logarithmic (**Note:** A linear scale evenly spaces values, where each increment represents a consistent change. A logarithmic scale uses multiples of the starting value, with each increment representing the same percentage increase.):

Figure 1.1a: <https://ourworldindata.org/grapher/historys-hockey-stick-worldwide-historical-gross-domestic-product-percapita-1990> (linear scale and ratio scale or logarithmic)

Figure 1.1b: <https://ourworldindata.org/grapher/historys-hockey-stick-gross-domestic-product-per-capita-using-the-ratio-scale-1990?yScale=linear> (linear scale and ratio scale or logarithmic)

- a) For Britain, identify a period of time when its growth rate was increasing and another period in which its growth rate was roughly constant. Which figure did you use, and why? [**Hint:** Choose between the Linear and Logarithmic scale in the "Axis scale" in "Settings" to observe the different effects]
- b) Identify a period during which GDP per capita was shrinking (a negative growth rate) faster than in India. Which figure did you use and why? [**Hint:** Choose between the Linear and Logarithmic scale in the "Axis scale" in "Settings" to observe the different effects]

See spreadsheet (Exercises_Week1_English.xlsx, sheets "Log scale" and "Log scale2" in Aula Global) for data.

2. **The *ceteris paribus* assumption in economic models:** Suppose you build a model of the market for umbrellas, in which the predicted number of umbrellas sold by a shop depends on their colour and price. In this model, we employ certain *ceteris paribus* assumptions, meaning that other variables are held constant.

- a) While the colour and the price are variables used to predict sales, which other variables do you think are being held constant?

Which of the following questions do you think this model might be able to answer? In each case, suggest improvements to the model that might help you to answer the question.

- a) Why are annual umbrella sales higher in the capital city than in other towns?
 - b) Why are annual umbrella sales higher in some shops in the capital city than others?
 - c) Why have weekly umbrella sales in the capital city risen over the last six months?
3. Download the spreadsheet data used to create Figure 1.2 that is in the text (<https://tzvetanmoev.github.io/core-skyscraper-1-income/>). Choose five countries and answer the following questions:
- a) Calculate their “Rich/poor ratio” for income in 1980, 2000 and 2020 (the ratio of the average income of the tenth (‘rich’) and first (‘poor’) deciles).
 - b) Describe the differences between countries and the changes over time that you find.
 - c) Can you think of any explanations for the observed changes or patterns?

[**Note:** The rich/poor ratio used here is similar to, but not exactly the same as, a commonly used measure of inequality called the 90/10 ratio. The 90/10 ratio is defined as the ratio between the income of the two individuals at the ninetieth and tenth percentiles. We are instead taking the ratio of the average income of the tenth (‘rich’) and first (‘poor’) deciles. The tenth decile is made up of all the people with higher income than the person at the ninetieth percentile, so its average is larger than the income of that person. The first decile is made of all the people with income less than the person at the tenth percentile, and so its average will be lower than the income of that person. Therefore, our rich/poor ratio will be a higher number than the 90/10 ratio for the same country.]

See spreadsheet “Exercises_Week1_English.xlsx”, sheet “World Inequality Database” in Aula Global for the data set.

2. MULTIPLE CHOICE QUESTIONS

4. The GDP per capita of Greece was \$22,494 in 2012 and \$21,966 in 2013. Based on these figures, the growth rate of GDP between 2012 and 2013 (to two decimal places) was:
- a) -2.40%
 - b) 2.35%
 - c) -2.35%
 - d) -0.24%
5. The following table shows the nominal GDP (in 2015 US dollars) and the population of Japan in 2013 and 2014 (source: The World Bank). Based on this information, which of the following statements regarding GDP per capita is correct?

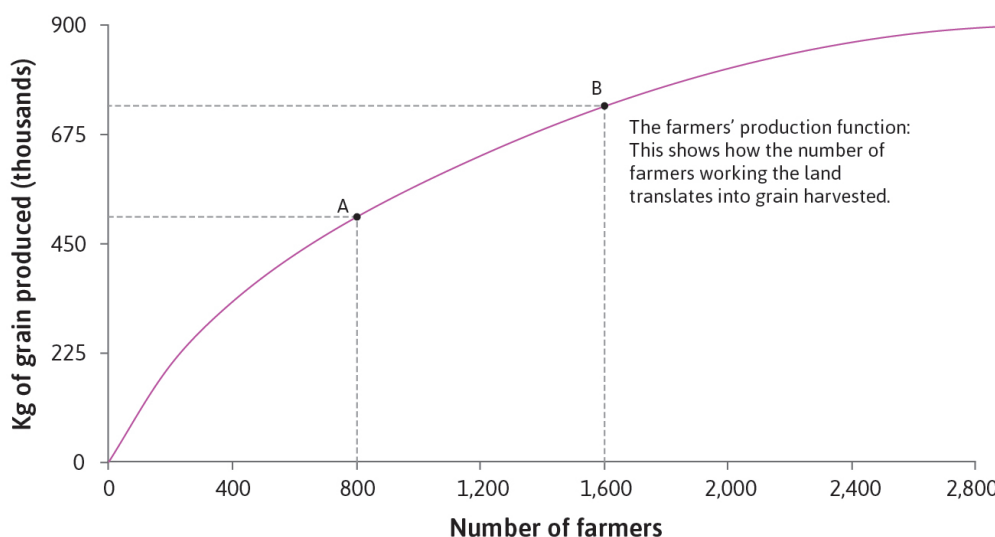
YEAR	2013	2014
GDP	\$4,919,563,108,372.50	\$4,601,461,206,888.10
POPULATION	127,338,621	127,131,800

- Japan's GDP per capita in 2013 was \$36,194.42.
- Japan's GDP per capita fell by 6.74% between 2013 and 2014.
- The fall in the population was enough to offset the fall in GDP, for an overall growth in GDP per capita between 2013 and 2014.
- Japan's GDP per capita fell by 6.31% between 2013 and 2014.

6. Which of the following are examples of markets?

- wartime food rationing
- auction websites such as eBay
- ttouts selling tickets outside concert halls
- sale of illegal arms

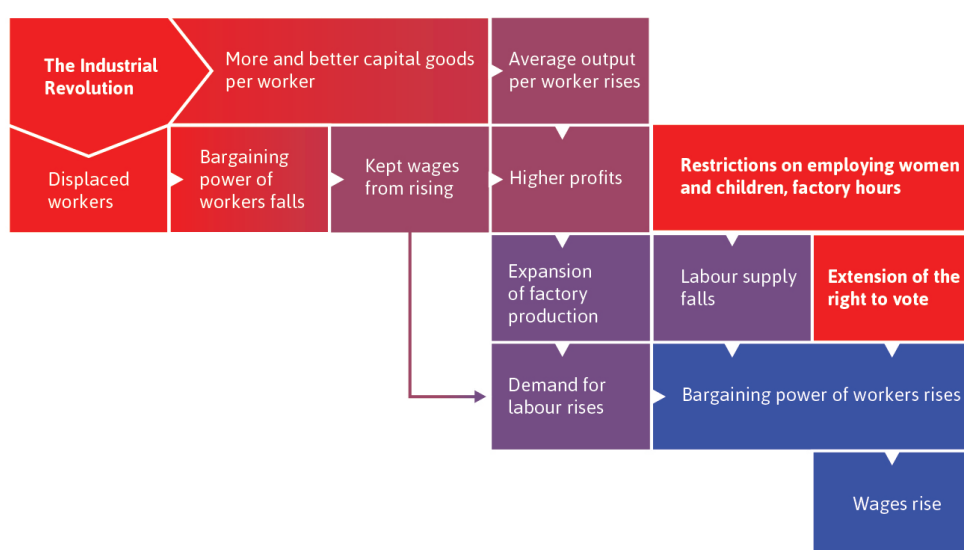
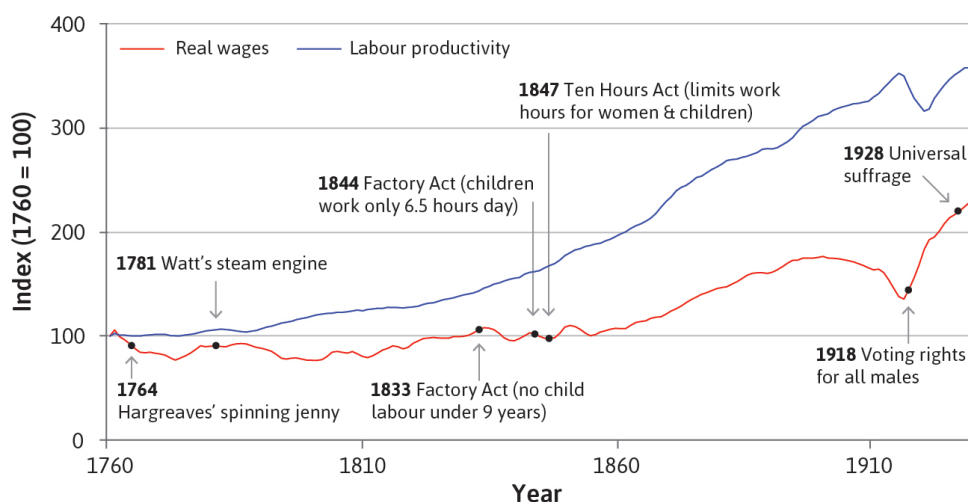
7. The following diagram depicts the production function of the farmers, where diminishing average product of labour is assumed. At A the average product of labour is $500,000/800 = 625$ kg of grain per farmer. At B the average product of labour is $732,000/1,600 = 458$ kg of grain per farmer. If you know that the grain output for 2,800 farmers is 894,000kg, then which of the following statements is correct?



See spreadsheet (Exercises_Week1_English.xlsx in Aula Global, sheet “Question 5 of Other Questions”) for table and graph.

- When the labour input is 2,800, the average product of labour is 300 kg.

- b. The slopes of the rays from the origin to the production function increase along the curve, indicating that the average product of labour is decreasing.
 - c. If the production function curve is an upward-sloping straight line, then there is no diminishing average product of labour.
 - d. It is possible that initially there are economies of scale (for example, when there are two farmers instead of one, the average output increases as they efficiently share the workload).
8. The following figures show the real wage index in England from 1750 to 2000, and the explanation of how England escaped the Malthusian trap in the period up to 1900. Based on this information, which of the following statements are correct
- a. These figures show evidence that increased productivity of labour increases wages immediately.
 - b. Throughout the 1800s, political developments led to a fall in the workers' bargaining power, keeping the real wage low.
 - c. England was able to escape from the Malthusian trap after the 1780s by being able to increase the workers' share of the rising output, due to an increase in the workers' bargaining power.
 - d. These figures demonstrate that the size of pie (the total output) does not determine the wage rate, but it merely sets a limit on how high the wage could possibly be.



3. NUMERICAL PROBLEMS

9. Suppose the price of one unit of labour (the wage) is 10 euros and the price of one unit of coal is 5 euros.
- What is the relative price of labour?
 - Using the method explained in the text, write down the equation of the isocost line for a cost of 60 euros, and write it in the standard form, $y = a + b x$.
 - Write down the isocost line in both forms when the cost is 30 euros and when the cost is 90 euros. Represent all the isocost lines in a graph. What happens to the isocost line when the price of coal rises to 20 euros and the wage remains equal to 10 euros? Compare the initial isocost line and the new isocost line using the same diagram for a total cost of 60 euros.

See spreadsheet (Exercises_Week1_English.xlsx, sheet "Problem 2" in Aula Global) for graphs.

4. CONCEPTUAL QUESTIONS

Write down a short and concise answer. When you are asked to solve the question in class, explain the concept clearly and give examples or pieces of evidence.

10. While campaigning for the US presidency on 18 March 1968, Senator Robert Kennedy gave a famous speech questioning ‘the mere accumulation of material things’ in American society. He said that GDP “counts the destruction of the redwood and the loss of our natural wonder in chaotic sprawl. It counts napalm and counts nuclear warheads and armoured cars for the police to fight the riots in our cities. It counts Whitman's rifle and Speck's knife, and the television programs which glorify violence in order to sell toys to our children. Yet the gross national product does not allow for the health of our children, the quality of their education or the joy of their play. It does not include the beauty of our poetry or the strength of our marriages, the intelligence of our public debate or the integrity of our public officials. It measures neither our wit nor our courage, neither our wisdom nor our learning, neither our compassion nor our devotion to our country, it measures everything in short, except that which makes life worthwhile. And it can tell us everything about America except why we are proud that we are Americans.” (Underline has been added). Please read the full text available in <https://www.jfklibrary.org/learn/about-jfk/the-kennedy-family/robert-f-kennedy/robert-f-kennedy-speeches/remarks-at-the-university-of-kansas-march-18-1968>.

What statements of Robert Kennedy do you agree with and what statements you don't agree with? For instance, do you think that GDP “counts the destruction of the redwood and the loss of our natural wonder...” and that GDP “...does not allow for the health of our children, the quality of their education or the joy of their play”?